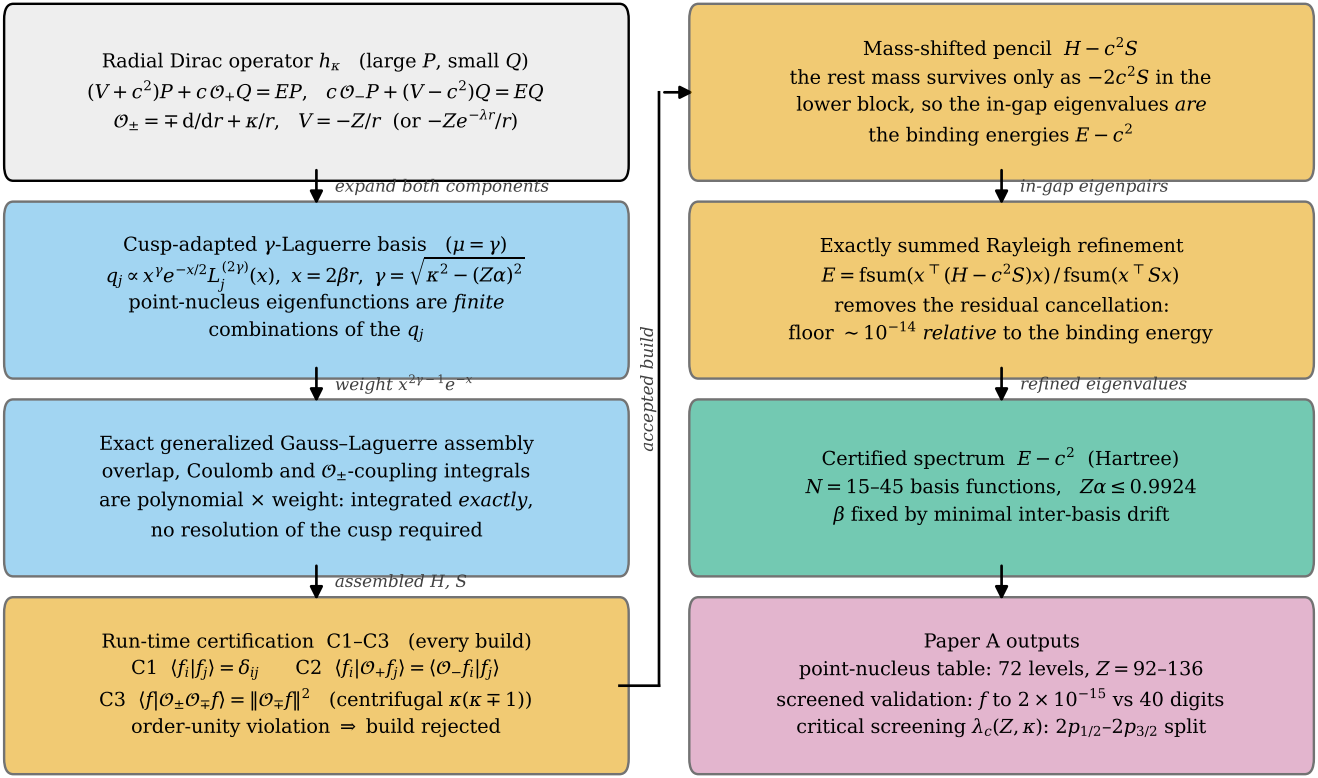
Radial Dirac operator h_κ (large P, small Q)
$(V + c^2)P + c\,\mathcal{O}_+ Q = EP$, $c\,\mathcal{O}_- P + (V - c^2)Q = EQ$
$\mathcal{O}_\pm = \mp\,\mathrm{d}/\mathrm{d}r + \kappa/r$, $V = -Z/r$ (or $-Ze^{-\lambda r}/r$)
expand both components
Cusp-adapted γ-Laguerre basis $(\mu = \gamma)$
$q_j \propto x^\gamma e^{-x/2} L_j^{(2\gamma)}(x)$, $x = 2\beta r$, $\gamma = \sqrt{\kappa^2 - (Z\alpha)^2}$
point-nucleus eigenfunctions are *finite* combinations of the q_j
weight $x^{2\gamma-1}e^{-x}$
Exact generalized Gauss–Laguerre assembly
overlap, Coulomb and $\mathcal{O}_\pm$-coupling integrals
are polynomial × weight: integrated *exactly*,
no resolution of the cusp required
assembled H, S
Run-time certification C1–C3 (every build)
C1 $\langle f_i | f_j \rangle = \delta_{ij}$ C2 $\langle f_i | \mathcal{O}_+ f_j \rangle = \langle \mathcal{O}_- f_i | f_j \rangle$
C3 $\langle f | \mathcal{O}_\pm \mathcal{O}_\mp f \rangle = \| \mathcal{O}_\mp f \|^2$ (centrifugal $\kappa(\kappa \mp 1)$)
order-unity violation ⇒ build rejected
accepted build
Mass-shifted pencil $H - c^2 S$
the rest mass survives only as $-2c^2 S$ in the
lower block, so the in-gap eigenvalues *are*
the binding energies $E - c^2$
in-gap eigenpairs
Exactly summed Rayleigh refinement
$E = \mathrm{fsum}(x^\top (H - c^2 S) x) / \mathrm{fsum}(x^\top S x)$
removes the residual cancellation:
floor $\sim 10^{-14}$ *relative* to the binding energy
refined eigenvalues
Certified spectrum $E - c^2$ (Hartree)
$N = 15$–45 basis functions, $Z\alpha \le 0.9924$
β fixed by minimal inter-basis drift
Paper A outputs
point-nucleus table: 72 levels, $Z = 92$–136
screened validation: f to 2×10^{-15} vs 40 digits
critical screening $\lambda_c(Z, \kappa)$: $2p_{1/2}$–$2p_{3/2}$ split